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SMART FACTORY EQUIPMENT MONITORING PLATFORM

SYSTEM OVERVIEW

PROBLEM STATEMENT

Manufacturing facilities rely on industrial equipment such as conveyor belts, robotic arms, pumps, and CNC machines to maintain production efficiency. Unexpected equipment failures can lead to downtime, reduced productivity, and increased maintenance costs.

PROJECT PURPOSE

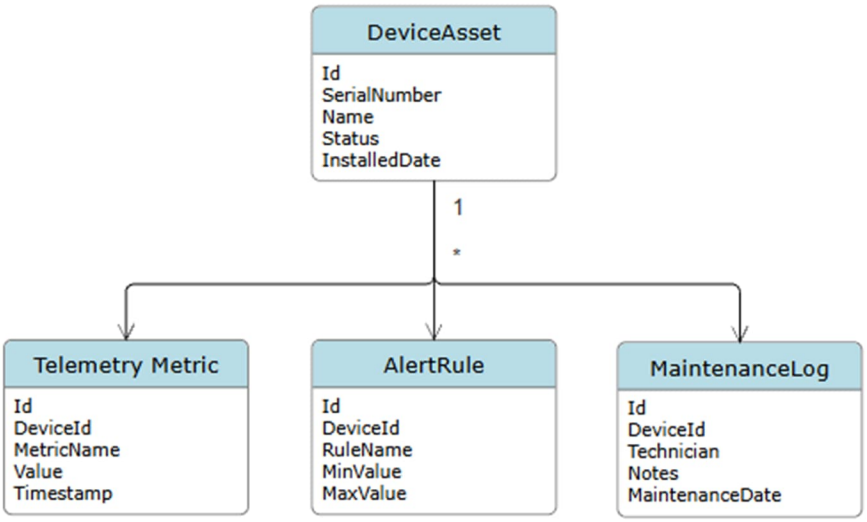
The Smart Factory Equipment Monitoring Platform is an Industrial Internet of Things (IIoT) solution designed to help factory employees manage equipment and monitor operational health through telemetric data collection. The platform tracks device status, evaluates telemetry readings against configurable alert thresholds, and records maintenance logs in order to support equipment reliability and predictive maintenance.

PLATFORM SCOPE

The platform provides the ability to:

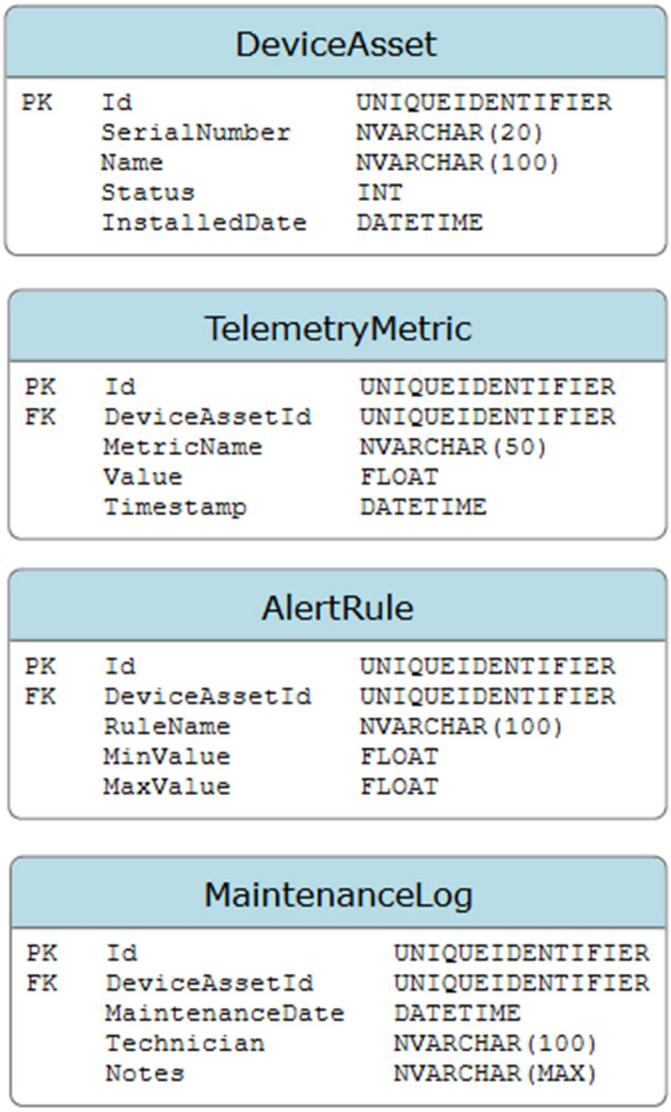
- Register and manage industrial equipment
- Collect telemetric measurements from connected devices
- Configure alert rules for equipment operating thresholds
- Record and manage maintenance activities
- Monitor equipment throughout its operational lifecycle

DOMAIN MODEL DIAGRAM



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ENTITY RELATIONSHIP DIAGRAM



BUSINESS RULES

SERIAL NUMBER VALIDATION

Device serial numbers must follow the following format:

DEV-12345

Invalid serial numbers generate a DomainException.

DEVICE STATUS LIFECYCLE

Valid transitions are as follows:

- Registered ➔ Active
- Active ➔ Maintenance
- Maintenance ➔ Active
- Active ➔ Retired
- Maintenance ➔ Retired

Invalid transitions generate a DomainException.

ALERT RULE OPERATING LIMITS

- Minimum values must be positive.
- Maximum values must be positive.
- Minimum values must be less than maximum values.

Invalid configurations generate a DomainException.

GITHUB LINK

https://github.com/freakyladymoonbeam/Capstone-Project